Níckolas de Aguiar Alves (he/him)

PhD candidate in physics
Federal University of ABC
Santo André, São Paulo, Brazil
Portuguese (native), English (fluent), Italian (basic)
Brazilian

aguiaralves.com
nick@aguiaralves.com
alves.nickolas@ufabc.edu.br

Education

2023– Sep.
Doctor in Physics
Federal University of ABC (UFABC), Santo André, São Paulo, Brazil
Advisor: André G. S. Landulfo

2021–2023 Feb. Apr.
Master in Physics
Federal University of ABC (UFABC), Santo André, São Paulo, Brazil
Advisor: André G. S. Landulfo
Thesis: *Nonperturbative Aspects of Quantum Field Theory in Curved Spacetime*

2017–2020 Feb. Dec.
Bachelor in Physics
Institute of Physics, University of São Paulo (IFUSP), São Paulo, São Paulo, Brazil
Advisor: João C. A. Barata

Academic Publications

Research Articles

1. **N. Aguiar Alves** and A. G. S. Landulfo. *Sound as a gauge theory and its infrared triangle*. 2026. arXiv: 2512.15796 [hep-th]. Submitted.
2. **N. Aguiar Alves** and A. G. S. Landulfo. "Null infinity as a Killing horizon". *Phys. Rev. D* **112**.6, 065017 (2025). arXiv: 2504.12514 [gr-qc].
3. **N. Aguiar Alves**, A. G. S. Landulfo, and B. A. Costa. "Positive mass in general relativity without energy conditions". *Phys. Rev. D* **111**.4, 044027 (2025). arXiv: 2408.00154 [gr-qc].

Review Articles

1. **N. Aguiar Alves**. "Lectures on the Bondi–Metzner–Sachs group and related topics in infrared physics". *Eur. Phys. J. C* **86**.5, 507 (2026). arXiv: 2504.12521 [gr-qc].

Pedagogical Articles

1. C. C. Rodrigues Evangelista and **N. Aguiar Alves**. "Using thermodynamics to learn gravitational wave physics". *Eur. J. Phys.* **47**.2, 025604 (2026). arXiv: 2602.21261 [gr-qc].

Essays and Short Contributions

1. **N. Aguiar Alves**. "Physics as a branch of poetry". *Am. J. Phys.* **94**.7 (2026), p. 507.
2. **N. Aguiar Alves** and B. A. Costa. *The Measure of a Mass*. 2025. arXiv: 2503.18963 [gr-qc]. Pre-published.

Theses

1. **N. Aguiar Alves**. "Nonperturbative Aspects of Quantum Field Theory in Curved Spacetime". MSc thesis. Santo André, Brazil: Federal University of ABC, 2023. xxiv, 152 pp. arXiv: 2305.17453 [gr-qc].

Other Works

1. **N. Aguiar Alves**. "Quantum Field Theory in Curved Spacetime. An Introduction". Unpublished lecture notes. 2023.

In Preparation

1. **N. Aguiar Alves**, A. G. S. Landulfo, and A. D. Pereira. *Nonperturbative renormalization group flow for a particle detector*. In preparation.
2. **N. Aguiar Alves**, C. C. Rodrigues Evangelista, G. J. Olmo, and D. Rubiera-Garcia. *Platypus stars: Exotic compact objects supported by vacuum pressure*. In preparation.

Funding

2026– PhD Scholarship – Research Internship Abroad
Explorations with Twistors and Cosmological Correlators
São Paulo Research Foundation (FAPESP) Grant No. 2026/02509-8
Advisor in Brazil: André G. S. Landulfo
Supervisor abroad: Guilherme L. Pimentel

2025– PhD Scholarship
The Sky as a Killing Horizon and Other Topics on the Infrared Structure of Gravity
São Paulo Research Foundation (FAPESP) Grant No. 2025/05161-0
Advisor: André G. S. Landulfo

2023–2025 PhD Scholarship
Coordenação de Aperfeiçoamento do Pessoal de Ensino Superior (CAPES)
Granted to pursue my PhD after placing first in UFABC's admission ranking

2021–2023 MSc Scholarship
The Functional Renormalization Group in Quantum Field Theory in Curved Spacetimes
São Paulo Research Foundation (FAPESP) Grant No. 2021/07372-7
Advisor: André G. S. Landulfo

2021 MSc Scholarship
Coordenação de Aperfeiçoamento do Pessoal de Ensino Superior (CAPES)
Granted to pursue my MSc after placing first in UFABC's admission ranking

2019–2020 BSc Scholarship
Hyperbolic Equations
São Paulo Research Foundation (FAPESP) Grant No. 2019/12158-4
Advisor: João C. A. Barata

Academic Service

Conferences Organized

2026 (scheduled) Thermality in Quantum Field Theory in Curved Spacetimes
Main organizer
Promoted by the ICTP South American Institute for Fundamental Research

2023 Golden Wedding of Black Holes and Thermodynamics
Main organizer

Schools Organized

2024– São Paulo School in Gravitational Physics
Main organizer
Part of ICTP Physics Without Frontiers (second edition onward)

Academic Service (continued)

2018–2021 — Jayme Tiomno School in Theoretical Physics
Founder and organizer of the first three editions

Peer Review

2026– — Reviewer for the American Journal of Physics

Scientific Societies

2025– — Brazilian Physical Society
2024– — International Society for Quantum Gravity

Teaching Experience

Minicourses and Workshops

2025 — Introduction to Black Hole Physics
II São Paulo School in Gravitational Physics
Minicourse for early undergraduate students

2024 — Infrared Symmetries of General Relativity
I São Paulo School in Gravitational Physics
Minicourse for advanced undergraduate and early graduate students

2023 — Quantum Field Theory in Curved Spacetime
Golden Wedding of Black Holes and Thermodynamics
Minicourse for early graduate students

2021 — Algebraic Methods of Theoretical Physics
III Jayme Tiomno School in Theoretical Physics
Minicourse on group theory in physics for first-year undergraduate students

2020 — An Introduction to LaTeX
XV Oceanography Thematic Week
Oceanographic Institute, University of São Paulo

2018 — An Introduction to LaTeX
I Jayme Tiomno School in Theoretical Physics

Teaching Assistant

2025 — Topics in Celestial Holography
Institute of Physics, University of São Paulo
Graduate-level course

Advanced Topics in General Relativity
Institute of Physics, University of São Paulo
Graduate-level course

2024 — Electromagnetic Phenomena
Federal University of ABC
Undergraduate-level course covering introductory electrodynamics

2021 — Classical Mechanics II
Federal University of ABC
Undergraduate-level course covering analytical mechanics

2019 — Mathematical Physics I
Institute of Physics, University of São Paulo
Undergraduate-level course covering introductory Fourier analysis

Teaching Experience (continued)

Participation in Panels

2026 — VI ICTP-SAIFR Summer School for Young Physicists
Panel with graduate students in physics

Outreach and Miscellanea

VI ICTP-SAIFR Summer School for Young Physicists
Tutored a course on black hole physics for high school students

2025 — International Young Physicists Tournament (Brazil)
Juror for the final stage of the 2025 edition of the IYPT Brazil

V ICTP-SAIFR Summer School for Young Physicists
Tutored a course on black hole physics for high school students

2018– — Physics Stack Exchange
Active contributor to the Physics Stack Exchange Q&A website
Most prolific contributor to the `qft-in-curved-spacetime` tag

Awards

2025 — "Outstanding Recognition" award at the UFABC Physics Week

- awarded to 12 out of 45 participants based on talks about their research progress

Research Presentations

Invited Talks

2026 — Sound as a gauge theory and its infrared triangle

- Simon's Collaboration on Celestial Holography Zoom Seminars (10 Mar. 2026)

2025 — The sky as a Killing horizon

- Siembra-HoLAGrav Future of Physics (08 May 2025)

Contributed Talks

2026 — Sound as a gauge theory and its infrared triangle

- Simons–Balseiro School and Workshop on Advanced Aspects of QFT (Instituto Balseiro, 18–27 May 2026)

2025 — Quantum Gravity and the Night Sky

- UFABC Physics Week (Federal University of ABC, 14–17 Oct. 2025)

2024 — Positive mass in general relativity without energy conditions

- XLVII Paulo Leal Ferreira Congress (Institute of Theoretical Physics, São Paulo State University, 29 Oct.–01 Nov. 2024)

2023 — Nonperturbative renormalization group flow for a particle detector

- Golden Wedding of Black Holes and Thermodynamics (online conference, 04–08 Dec. 2023)

Poster Presentations

2026 — We can use the renormalization group to study the nonperturbative behavior of particle detectors

- Simons–Balseiro School and Workshop on Advanced Aspects of QFT (Instituto Balseiro, 18–27 May 2026)

Research Presentations (continued)

2025 — Linear sound has asymptotic symmetries

- XI QCD Meets Gravity Conference (Principia Institute, 15–19 Dec. 2025)
- XI QCD Meets Gravity School (Institute of Theoretical Physics, São Paulo State University, 08–12 Dec. 2025)

Stars with extreme pressures can mimic black holes

- XLVIII Paulo Leal Ferreira Congress (Institute of Theoretical Physics, São Paulo State University, 14–17 Oct. 2025)

Killing horizons naturally define asymptotic symmetries

- First International Latin American Conference on Gravitational Waves (National Institute for Space Research, 15–19 Sep. 2025)
- VI Siembra-HoLAGrav Young Frontiers Meeting at ICTP-SAIFR (Institute of Theoretical Physics, São Paulo State University, 30 June–11 July 2025)
- Program on Anomalies, Topology and Quantum Information in Field Theory and Condensed Matter Physics (Principia Institute, 16–27 June 2025)

2024 — Negative masses are unstable and we don't need energy conditions to prove it

- Witnessing Quantum Aspects of Gravity in a Lab (Principia Institute, 23–27 Sep. 2024)

2023 — Nonperturbative aspects of quantum field theory in curved spacetime

- XLVI Paulo Leal Ferreira Congress (Institute of Theoretical Physics, São Paulo State University, 24–27 Oct. 2023)
- V Jayme Tiomno Physics School (Institute of Physics, University of São Paulo, 4–8 Sep. 2023)
- Interfaces Between Quantum and Classical Statistical Mechanics (Institute of Mathematics and Statistics, University of São Paulo, 24–28 July 2023)

Flash Talks

2023–2024 — Nonperturbative renormalization group flow for a particle detector

- School on Phase Transitions and Gravitational Waves (International Institute of Physics, Federal University of Rio Grande do Norte, 4–8 Mar. 2024)
- Quantum Spacetime and the Renormalization Group 2023 (Sant'Elmo Beach Hotel, 2–6 Oct. 2023)